AI PRODUCT MANAGEMENT 2 · DETAILED COURSE

# Retrieval-Augmented Generation (RAG)

Grounding model answers in your own data: architecture, retrieval quality, evaluation and production operation.

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CHAPTER 14

# Retrieval-Augmented Generation (RAG)

Grounding model answers in your own data: architecture, retrieval quality, evaluation and production operation.

## RAG Fundamentals

**Retrieve relevant documents at request time and give them to the model to ground its answer.**

Retrieval-augmented generation finds relevant documents at request time and gives them to the model so its answer is grounded in them. It lets a product answer from current company data without retraining the model.

## Why RAG Is Required

**Models lack your private or recent data and can hallucinate; RAG supplies sources and enables citations.**

Models do not know your private or recent data and can hallucinate. RAG supplies sources, which enables citations and makes answers checkable.

## RAG vs Fine-Tuning

**RAG adds knowledge at query time and is easy to update; fine-tuning changes behaviour or style. They are often combined.**

RAG adds knowledge at query time and is easy to update. Fine-tuning changes behaviour, style or format. Many products start with RAG and fine-tune only if behaviour still falls short.

## RAG Architecture

**Ingest and index, retrieve, rerank, build context, generate, cite, and monitor.**

The pipeline ingests and indexes documents, retrieves candidates for a query, reranks them, builds the context, generates the answer with citations and is monitored for quality.

## Data Ingestion

**Collecting documents from sources and keeping the index in sync with changes.**

Ingestion collects documents from sources such as drives, wikis and databases, then keeps the index in sync when content changes. Stale content is a common cause of wrong answers.

## Document Parsing

**Extracting clean text, tables and structure from PDFs, web pages and office files.**

Parsing extracts clean text, tables and headings from PDFs, web pages and office files. Poor parsing silently degrades every later step, so check parsed output by hand.

### Data Cleaning

**Removing noise, duplicates and boilerplate and fixing encoding so retrieval is accurate.**

Cleaning removes boilerplate, duplicates and noise, and fixes encoding problems. Clean data makes retrieval more accurate and smaller.

### Document Chunking

**Splitting documents into passages sized for retrieval and context limits.**

Chunking splits documents into passages sized for retrieval and the context window. Chunks that are too small lose meaning; chunks that are too large dilute relevance.

### Chunking Strategies

**Fixed size, recursive, by heading or semantic boundaries, with overlap to preserve context.**

Options include fixed-size chunks, recursive splitting, splitting by heading, and semantic boundaries, often with some overlap to keep context. Evaluate each option on your own questions.

### Embedding Models

**Models that convert text to vectors; choose by quality, language coverage, dimensions and cost.**

Embedding models turn text into vectors. Compare them on retrieval quality for your language and domain, plus dimensions, cost and speed.

### Vector Databases

**Stores optimised for nearest-neighbour search over embeddings.**

Vector databases store embeddings and run fast nearest-neighbour search. They also support metadata filters. Choose based on scale, hosting needs and integration.

### Pinecone

**A managed vector database service with simple scaling.**

Pinecone is a managed vector database service with simple scaling. Teams that do not want to run infrastructure often choose a managed option.

### Weaviate

**An open-source vector database with hybrid search and modules.**

Weaviate is an open-source vector database with hybrid search and modules for embedding. It can be self-hosted or used as a managed service.

### Chroma

**A lightweight open-source vector store popular for prototypes.**

Chroma is a lightweight open-source vector store, popular for prototypes and small applications. Check its production readiness for your scale before committing.

### FAISS

**Meta's library for efficient similarity search over vectors, run in your own process.**

FAISS is a library from Meta for efficient similarity search, run inside your own process. It is fast but does not provide a full database service by itself.

### Semantic Search

**Finding passages by meaning using embeddings, even without shared keywords.**

Semantic search finds passages by meaning, even when they do not share keywords. It handles paraphrases well but can miss exact codes and names.

### Keyword Search

**Matching exact terms (BM25), strong for names, codes and rare words.**

Keyword search, often BM25, matches exact terms. It is strong for names, product codes and rare words that embeddings can blur.

### Hybrid Search

**Combining keyword and semantic scores for better recall.**

Hybrid search combines keyword and semantic scores. It usually improves recall over either method alone, at the cost of more tuning.

### Metadata Filtering

**Restricting retrieval by attributes such as date, source, permission or product.**

Filters restrict retrieval by attributes such as date, source, product or user permission. Permission filters are essential so users only retrieve documents they may read.

### Retrieval Strategies

**Top-k, multi-query, parent-document, query rewriting and filters tuned to the content.**

Options include top-k retrieval, multi-query retrieval, parent-document retrieval and query rewriting. Choose based on the failure patterns you see in evaluation.

### Reranking

**A second model reorders retrieved candidates by relevance before generation.**

A reranker, a second model, reorders retrieved candidates by relevance before generation. It improves precision at the cost of extra latency.

### Context Construction

**Selecting, ordering and formatting retrieved passages inside the prompt.**

Context construction selects, orders and formats retrieved passages inside the prompt, with source labels. Include only what supports the answer, since extra text can distract the model.

### Grounded Generation

**Instructing the model to answer only from provided sources and say when they are insufficient.**

The model is instructed to answer only from the provided sources and to say when they are insufficient. Test that it does this on questions the documents do not cover.

### Citation and Source Attribution

**Linking each claim to its source so users can verify.**

Each claim links to its source so users can verify it. Citations build trust, but they must point to the passage that actually supports the claim.

### RAG Evaluation

**Measuring retrieval and answer quality separately on a labelled question set.**

Evaluate retrieval and answer quality separately on a labelled set of questions. A wrong answer may come from retrieval or from generation, and you need to know which.

### Retrieval Precision and Recall

**Precision is the share of retrieved passages that are relevant; recall is the share of relevant passages retrieved.**

Precision is the share of retrieved passages that are relevant. Recall is the share of relevant passages that were retrieved. Both must be measured to tune chunking and ranking.

### Faithfulness and Context Relevance

**Faithfulness checks the answer is supported by the context; context relevance checks the context fits the question.**

Faithfulness checks the answer is supported by the context. Context relevance checks the retrieved context fits the question. Both can be scored with a model-based judge validated against human labels.

### RAG Performance Optimisation

**Tuning chunking, embeddings, hybrid search, reranking, caching and prompts for quality, speed and cost.**

Tune chunking, embeddings, hybrid search, reranking, caching and prompts for quality, speed and cost. Change one factor at a time and re-run the evaluation.

### Production RAG Monitoring

**Tracking retrieval quality, unanswered questions, latency, cost and freshness in live use.**

Monitor retrieval quality, unanswered questions, latency, cost and document freshness in live use. Questions with no good source are a useful signal for content gaps.

## Summary of terms

| Term | Meaning |
| --- | --- |

| | |
|---|---|
| **RAG Fundamentals** | Retrieve relevant documents at request time and give them to the model to ground its answer. |
| **Why RAG Is Required** | Models lack your private or recent data and can hallucinate; RAG supplies sources and enables citations. |
| **RAG vs Fine-Tuning** | RAG adds knowledge at query time and is easy to update; fine-tuning changes behaviour or style. They are often combined. |
| **RAG Architecture** | Ingest and index, retrieve, rerank, build context, generate, cite, and monitor. |
| **Data Ingestion** | Collecting documents from sources and keeping the index in sync with changes. |
| **Document Parsing** | Extracting clean text, tables and structure from PDFs, web pages and office files. |
| **Data Cleaning** | Removing noise, duplicates and boilerplate and fixing encoding so retrieval is accurate. |
| **Document Chunking** | Splitting documents into passages sized for retrieval and context limits. |
| **Chunking Strategies** | Fixed size, recursive, by heading or semantic boundaries, with overlap to preserve context. |
| **Embedding Models** | Models that convert text to vectors; choose by quality, language coverage, dimensions and cost. |
| **Vector Databases** | Stores optimised for nearest-neighbour search over embeddings. |
| **Pinecone** | A managed vector database service with simple scaling. |
| **Weaviate** | An open-source vector database with hybrid search and modules. |
| **Chroma** | A lightweight open-source vector store popular for prototypes. |
| **FAISS** | Meta's library for efficient similarity search over vectors, run in your own process. |
| **Semantic Search** | Finding passages by meaning using embeddings, even without shared keywords. |
| **Keyword Search** | Matching exact terms (BM25), strong for names, codes and rare words. |
| **Hybrid Search** | Combining keyword and semantic scores for better recall. |
| **Metadata Filtering** | Restricting retrieval by attributes such as date, source, permission or product. |
| **Retrieval Strategies** | Top-k, multi-query, parent-document, query rewriting and filters tuned to the content. |
| **Reranking** | A second model reorders retrieved candidates by relevance before generation. |
| **Context Construction** | Selecting, ordering and formatting retrieved passages inside the prompt. |
| **Grounded Generation** | Instructing the model to answer only from provided sources and say when they are insufficient. |
| **Citation and Source Attribution** | Linking each claim to its source so users can verify. |
| **RAG Evaluation** | Measuring retrieval and answer quality separately on a labelled question set. |
| **Retrieval Precision and Recall** | Precision is the share of retrieved passages that are relevant; recall is the share of relevant passages retrieved. |
| **Faithfulness and Context Relevance** | Faithfulness checks the answer is supported by the context; context relevance checks the context fits the question. |

| | |
|---|---|
| **RAG Performance Optimisation** | Tuning chunking, embeddings, hybrid search, reranking, caching and prompts for quality, speed and cost. |
| **Production RAG Monitoring** | Tracking retrieval quality, unanswered questions, latency, cost and freshness in live use. |

**Practice exercise:** Build a small RAG assistant over ten documents and measure retrieval precision and recall on twenty questions.

**Self-check**

1. Explain RAG Fundamentals in two sentences to a non-technical stakeholder, then say what you would measure for it.
2. Explain FAISS in two sentences to a non-technical stakeholder, then say what you would measure for it.
3. Explain Production RAG Monitoring in two sentences to a non-technical stakeholder, then say what you would measure for it.

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ORIGINAL CONTENT

# Original AI Product Management content

The AI Product Management programme already on SkillNavora, included unchanged so the full course is in one place.

**Quick links**Study Library (all material) Learning sites for PM AI builder tools (how-to) Webhooks & Playwright MCP AI Skills Academy AI Skills Map Context engineering (PDF) AI skills & templates (PDF) Orchestration (PDF) PRD library (120 ideas) AI PRD Writer AI PM Toolkit AI Evals (PDF) Browser automation (PDF) AI second brain (PDF) PM handbook Top 100 PM questions AI product sense Q&A (86) PM practice test Practice Lab Tool tutorials AI Agents & MCP for PMs

PRD documents

## PRD library: template, framing guide and 120 PRD ideas

Learn how top PMs frame a PRD, download the free template, and turn any of 120 ideas across 15 industries into a full PRD with AI.

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Context window

## Context window: what a model can see at once

The context window is the most tokens a model can consider in one request. It covers your prompt, any documents you paste, the conversation so far and the reply. Anything outside it is invisible to the model. For PMs: a longer window fits more material but costs more and can slow responses; put the most important facts near the instructions; use retrieval (RAG) rather than pasting whole libraries; and test answers on long inputs, where quality can drop.

Context engineering course Context windows in the AI PM 2 course

Tokens and context in the Grokbot course

AI-First Product Management

# Resources for Product Managers

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2Product Strategy and Strategic Thinking*Handbook chapter, detailed PDF, handbook and class slides*PDF Detailed PDF Handbook Class Slides
3Problem Discovery and Market Research*Handbook chapter, detailed PDF, handbook and class slides*PDF Detailed PDF Handbook Class Slides
4Problem Framing and Prioritization*Handbook chapter, detailed PDF, handbook and class slides*PDF Detailed PDF Handbook Class Slides
5UX and Design Collaboration*Handbook chapter, detailed PDF, handbook and class slides*PDF Detailed PDF Handbook Class Slides
6Effective Communication and Stakeholder Management*Handbook chapter, detailed PDF, handbook and class slides*PDF Detailed PDF Handbook Class Slides
7Product Analytics and Metrics*Handbook chapter, detailed PDF, handbook and class slides*PDF Detailed PDF Handbook Class Slides
8Product Launch, Growth, and Monetization*Handbook chapter, detailed PDF, handbook and class slides*PDF Detailed PDF Handbook Class Slides
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### AI Tools Lab

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### Business AI Stack (chapters 24 to 31)

24The business AI stack: tools and where to learn themPDF ↗ 25Next.js: customer-facing dashboardsPDF ↗ 26Zoho CRM: managing customers and leadsPDF ↗ 27WhatsApp Business Platform: customer messagingPDF ↗ 28Deployment with Vercel and AWSPDF ↗ 29Jira: tracking product and development workPDF ↗ 30Meta Business Suite, social media marketing and Google AdsPDF ↗ 31A one-page website that generates leadsPDF ↗

### Complete AI tool guides (15)

1n8n: workflows, Gmail and an AI reply agentPDF ↗ 2Make: scenarios, routers and lead routingPDF ↗ 3Zapier: invoices from Gmail to Drive and a sheetPDF ↗ 4Webhooks: how apps talk in real timePDF ↗ 5Replit: one app for the web and a phonePDF ↗ 6Lovable: a job application trackerPDF ↗ 7Figma: components, auto layout and handoffPDF ↗ 8Claude Code: an AI coding agent in your projectPDF ↗ 9MCP: connecting AI assistants to your toolsPDF ↗ 10LLMs: how they work and how to call themPDF ↗ 11RAG: answers from your own documentsPDF ↗ 12AI agents: a job-search agent with approvalsPDF ↗ 13Agentic AI: planning, guardrails and teamsPDF ↗ 14Generative AI: uses, limits and a weekly reportPDF ↗ 15Other AI tools: choosing one per jobPDF ↗

Complete AI tool guides (one PDF, 30 pages) ↗

## Libraries

★Framework Library*29 frameworks with worked examples and templates*PDF ↗ ★Case Study Library*12 product case studies*PDF ↗ ★Practical Projects and Portfolio*5 end-to-end portfolio projects*PDF ↗ ★Interview Handbook*100+ PM interview questions with model answers*PDF ↗ ★AI Tools Lab*Every AI tools class with step-by-step labs*PDF ↗

## PM interview practice (free, on the original sites)

Hand-picked public question banks and guides from other PM communities. Each link opens the original page in a new tab.

## PM frameworks cheat sheet

| Framework | Use it to | How it works |
|---|---|---|
| **RICE** | Prioritise a backlog | Reach × Impact × Confidence ÷ Effort. Score each idea, then sanity-check the ranking. |
| **Kano** | Decide which features delight | Classify features as Must-be, Performance, Attractive, Indifferent or Reverse using a short survey. |
| **MoSCoW** | Scope a release | Must have, Should have, Could have, Won't have (this time). |
| **Jobs-to-be-done** | Understand motivation | When [situation], I want to [motivation], so I can [outcome]. |
| **AARRR (Pirate metrics)** | Map the growth funnel | Acquisition, Activation, Retention, Referral, Revenue. |
| **HEART** | Measure user experience | Happiness, Engagement, Adoption, Retention, Task success, each with goals, signals and metrics. |
| **North Star metric** | Align the team | One metric that captures the value users get, supported by input metrics the team can move. |
| **Five whys** | Find the root cause | Ask 'why?' repeatedly until you reach a cause you can act on. |
| **CIRCLES** | Answer product design questions | Comprehend, Identify users, Report needs, Cut through prioritisation, List solutions, Evaluate trade-offs, Summarise. |

## PRD template

PRD: [Feature name] Owner: [Name] · Status: Draft · Last updated: [Date] 1. Problem Who has the problem, what it is, and the evidence (data, interviews, tickets). 2. Goal and success metrics Primary metric (with baseline and target), secondary metrics, guardrails. 3. Users and use cases Target segment, top 3 user stories: "As a [user], I want [action] so that [benefit]." 4. Solution What we'll build (link designs), what's out of scope, open questions. 5. AI behaviour (if applicable) Inputs, model, prompt/approach, what it must never do, human review, evaluation set and pass criteria, cost per request. 6. Launch plan Experiment design, rollout stages, support and comms, rollback criteria. 7. Risks and dependencies Technical, legal/privacy, operational; owners and mitigations.

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